



Market Technicians Association, Inc.

Professionals Managing Market Risk • Incorporated in 1973

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Chartered Market Technician (CMT) Program – Level II

The CMT Level II exam measures the candidates' competency in the **application** of concepts, theory, and techniques covered by the required readings. CMT Level II candidates must demonstrate their ability to apply concepts identified in their Level I studies to relevant conditions or scenarios.

Exam time length: 4 hours, 15 minutes

Exam format: Multiple Choice

The curriculum is organized into exam specific knowledge domains that provide a framework for recognizing and implementing investment/trading decisions. CMT Level II exam tests the candidate's knowledge in 12 domains:

1. Theory and History
2. Market Indicators
3. Construction
4. Trend Analysis
5. Chart and Pattern Analysis
6. Confirmation
7. Cycles
8. Selection and Decision
9. System Testing
10. Risk Management
11. Statistical Analysis
12. Ethics



CMT Level II Exam Topics & Question Weightings

1. Theory and History	a. behavioral finance b. Adaptive Market Hypothesis	5%	7
2. Market Indicators	a. breadth indicators b. sentiment measures c. volatility	8%	13
3. Construction	a. volume	3%	4
4. Trend Analysis	a. trendlines b. multiple time frame analysis c. breakouts d. moving averages e. trend strength indicators	15%	23
5. Chart and Pattern Analysis	a. gap analysis b. support and resistance	15%	23
6. Confirmation	a. oscillators and divergence b. sector rotation c. intermarket signals	6%	9
7. Cycles	a. seasonal cycles	3%	4
8. Selection and Decision	a. uncorrelated assets b. relative strength c. forecasting techniques	10%	15
9. System Testing	a. algorithmic development b. objective analysis of rules c. performance measures	10%	15
10. Risk Management	a. absolute and relative risk b. risk modeling c. value at risk d. volatility risk e. liquidity risk f. diversification g. leverage risk h. portfolio risk management i. risk-based performance measures	15%	23
11. Statistical Analysis	a. inferential statistics	7%	11
12. Ethics	a. Standards and Practices	3%	4

CMT Level II Exam - Learning Objectives

1. Theory and History	
Behavioral finance	Recognize evidence of cognitive biases. Identify cognitive biases in investment selection.
Adaptive Market Hypothesis	Contrast Efficient Market Hypothesis with Adaptive Market Hypothesis.
2. Market Indicators	
breadth indicators (e.g., A/D, up/down volume)	Interpret data and charts of market breadth indicators Recognize changes in market breadth and identify their significance
index construction	Distinguish between different methods for constructing a market or sector index Recognize the influence of index construction on price action
sentiment measures (e.g., put-call ratio, investor polls)	Contrast the uses of differing sentiment measures
volatility (e.g., vix, historical, implied)	Identify different measures of volatility Interpret volatility signals as part of a market forecast Compare volatility behavior with corresponding price behavior
3. Construction	
volume	Interpret volume data Analyze the behavior of a given volume-weighted indicator
4. Trend Analysis	
trend lines	Select valid trend lines Interpret the significance of trend line breaks
multiple time frame analysis	Compare trend signals over multiple time frames Identify evidence of changing trends in multiple time frames
breakouts (from channels or chart patterns)	Analyze breakout signals for use in forecasting Recognize evidence for improving confidence in breakout signals
moving averages	Contrast the use of various moving averages Analyze changes in moving average behavior Interpret signals given by various moving averages
trend strength indicators (e.g., DMI, ADX, etc.)	Determine the strength of a trend based on indicator data Select the correct definition of trend strength indicators
5. Chart and Pattern Analysis	
gap analysis	Recognize gap signals Evaluate the strength of various gap signals Classify gap types Identify support and resistance on given charts
support and resistance	Evaluate support and resistance evidence from data and charts for use in forecasts
6. Confirmation	
oscillators and divergence	Identify confirming divergence signals within oscillators
sector rotation	Recognize confirmation signals given from sector rotation data

Intermarket signals	Recognize confirmation signals inferred from intermarket analysis
7. Cycles	
seasonal cycles	Identify potential opportunity and risk based on seasonal cycle information Define methods for applying cycle studies
8. Selection and Decision	
uncorrelated assets	Determine appropriate asset selections based on correlation data
relative strength	Determine appropriate asset selections based on relative strength
forecasting techniques (pattern and trend recognition)	Determine appropriate asset selections based on trend and pattern forecasts
9. System Testing	
algorithmic development	Select correct procedures for proper development of algorithms Identify valid data output for algorithmic system testing
optimizing entry and exit rules (filtering)	Determine proper optimizing and filtering procedures for system testing
equity curve analysis	Identify valid system adjustments based on equity curve analysis
position size rules (e.g., Tharp's methods, Kelley criterion, Optimal f)	Recognize the influence of position size rules
profit measures (e.g., profit factor, outlier-adjusted profit to loss, percentage of winning trades, annualized rate of return, payoff ratio, length of average winning trade, efficiency factor)	Distinguish between different profit measures (profit factor, outlier-adjusted profit to loss ratio and others)
10. Risk Management	
absolute and relative risk (i.e. total risk v. risk compared to benchmark)	Determine differences in risk measures (absolute vs. relative, etc.)
risk modeling	Select appropriate risk modeling steps
value at risk	Identify appropriate use of Value at Risk (VaR)
volatility risk	Identify effective measures of volatility risk Identify volatility risk from given charts and data
liquidity risk	Select appropriate responses to liquidity risk
diversification	Select appropriate diversification strategies to mitigate risk
stops v. hedging	
leverage risk	Explain leverage risk for various asset classes

portfolio risk management (e.g., market neutral, relative strength)	Determine appropriate rules useful in portfolio risk management
risk measures (e.g., maximum cumulative drawdown, net profit to drawdown, maximum consecutive losses, largest losses, longest flat time, time to recovery, maximum + and - excursions)	Select appropriate risk measures for various objectives
risk-based performance measures	Define various risk-based performance measures (maximum cumulative drawdown, net profit to drawdown, maximum consecutive losses, largest losses, longest flat time, time to recovery, maximum favorable and adverse excursions)
11. Statistical Analysis	
inferential statistics (e.g., correlation, regression, t-test)	Identify proper application of inferential statistics methods in system development and testing Determine results from an analysis of correlation data Interpret results from regression or t-test data Analyze data from tests using inferential statistics
12. Ethics	<u>Code of Ethics and Standards of Professional Conduct</u>

